The Direct and Indirect Effects of Genetics and Education

Senan Hogan-Hennessy*
Economics Department, Cornell University†

This version: 26 June 2025

Job Market Paper, with newest version available here.

Abstract

Social science genetics has shown genes matter for education and labour market outcomes, but has little credible evidence on the economic mechanisms beneath these findings. I use plausibly random deviation from parents as a natural experiment for the Education PolyGenic Index (Ed PGI) to estimate the causal effect of genes associated with years of education, with data from the UK Biobank. I then decompose the total genetic genetic effects into a direct genetic effect and indirect education pathway using a causal mediation framework that allows for selection-into-education, using multiple UK university openings in the 1960s as instruments for higher education attendance. The Ed PGI increases labour market earnings through higher education attendance, with direct genetic effects indistinguishable from zero — contradicting previous speculation that education-linked genes independently increase intelligence or other labour market traits. Education linked genes are more about returns to higher education, and less about direct genetic effects. These findings reframe the usual focus in social science genetics by shifting from asking "if genes matter" to "how genes matter" — not as deterministic forces, but as factors that influence human capital decisions within institutional constraints. By using causal methods that account for individual choice in education, and not than assuming genetic determinism, this research provides a richer understanding of how biological inheritance, human capital decisions, and institutional structures interact to shape labour market inequality.

Keywords: Education, Mediation, Polygenic indicies, Nature/nurture.

JEL Codes: D31, D91, I24, J24, Z00.

---

*UK Biobank Project #335854. This work has been supported by research grants from the Centre for the Study of Inequality and the Industrial Labour Relations School, Cornell University. For helpful comments I thank Ryan Dycus, Benny Goldman, Takuma Habu, Jake Meyer, Douglas Miller, Zhuan Pei, and Evan Riehl. I thank seminar participants at EALE Bergen NHH (2024), and Cornell University (2023, 2024, 2025) for helpful discussion. Any comments or suggestions may be sent to me at seh325@cornell.edu.

†Address: Uris Hall #447, Economics Department, Cornell University NY 14853 USA.

> The difference of natural talents in different men, is, in reality, much less than we are aware of [...]. The difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise not so much from nature, as from habit, custom, and education.
>
> — *Smith (1776), The Wealth of Nations, Volume 1 p. 16.*

0.1 Writing Plan

Section map for this paper:

1. Intro, written last following https://blogs.ubc.ca/khead/research/research-advice/formula.
2. Data + biological context.
3. Section on the adjusted PGI randomisation design, clearly stating estimands
4. New results of genetic effects.
5. Conceptual framework for mediation effects, clearly stating estimands
6. Results of the mediation analysis.
7. Conclusion.

Note: a major concern for the Ed PGI previous papers is that they have little economic insight. This paper is framed as changing that, investigating the economic mechanisms underlying the Ed PGI causal effects, giving economic credence where before there was merely concern.

1 Introduction

This paper proceeds as follows. Section 2 summarises the UKB data, and explains how the Ed PGI is constructed. Section 3 shows a causal research design, based on Mendelian independent inheritance, for estimating the causal effects of the Ed PGI; Section 5 shows estimates of the causal effects of the Ed PGI, on education and labour market outcomes. Section 6 gives a conceptual framework for estimating the direct and indirect effects of genetics and education, in this observational setting; **??** gives the empirical results. Section 8 concludes.

2 Data and Genetic Context

2.1 UK Biobank (UKB)

The UK Biobank (UKB) is a large-scale prospective cohort study comprising approximately half a million members of the British public, recruited between 2006 and 2010. Letters were sent to a random sample of addresses in the National Health Service registry, and the final UKB data represent voluntary respondents.[1] Participants, aged 40-69 years at recruitment, underwent comprehensive baseline assessments, asked many questions for socioeconomic information either by interview, automated survey answers, or follow-up sessions. Key socioeconomic variables include educational attainment (measured in education qualifications), household income brackets, employment status, and industry of employment. See Sudlow et al. (2015) for a full description of the UKB organisation and data.

The UKB collected detailed information on educational qualifications, including completion of primary school, GCSEs (typically completed at age 16), A-Levels (completed at age 18), and completion of an higher education degree. I convert these qualifications into education years based on each participant's highest reported qualification, following the International Standard Classification of Education.[2] While the UKB collected household income data, this information was recorded only in categorical bins, limiting its utility for continuous analysis. Following Kweon et al. (2025), I construct an individual-level income measure using participants' 4-digit Standard Occupational Classification (SOC) codes. Specifically, I impute occupation-based wages by estimating hourly earnings for each SOC category using data from the British Annual Survey of Hours and Earnings, conditional on age, gender, and year. This approach provides a more granular and continuous measure of economic status than the categorical household income variable (see Appendix A.4 for details on the imputation procedure). I create the annual occupation income by multiplying this by the number of hours they reported working, times 52 weeks in the year. For an example, this procedure gives a female participant, age 58 in 2010 with SOC 2314 (i.e., a secondary education teacher by profession), an occupation hourly wage of £29.20 and an annual income of £68,476 for her 45 hours worked each week (adjusted to 2024 £GBP by CPI-H).

[1] The UKB may not be representative of the entire UK population, as it only represents willing respondents. As such, the findings of this paper should only be taken as internally valid among the UKB respondents, and the sibling analysis sample described.

[2] As noted by Okbay et al. (2022), British vocational degrees inconsistently coded by the International Standard Classification of Education. I follow the literature by assigning education years as the age they reported leaving full-time education minus five, among those with a vocational degree as their highest qualification.

Table 1: Descriptive Statistics, UK Biobank (UKB).

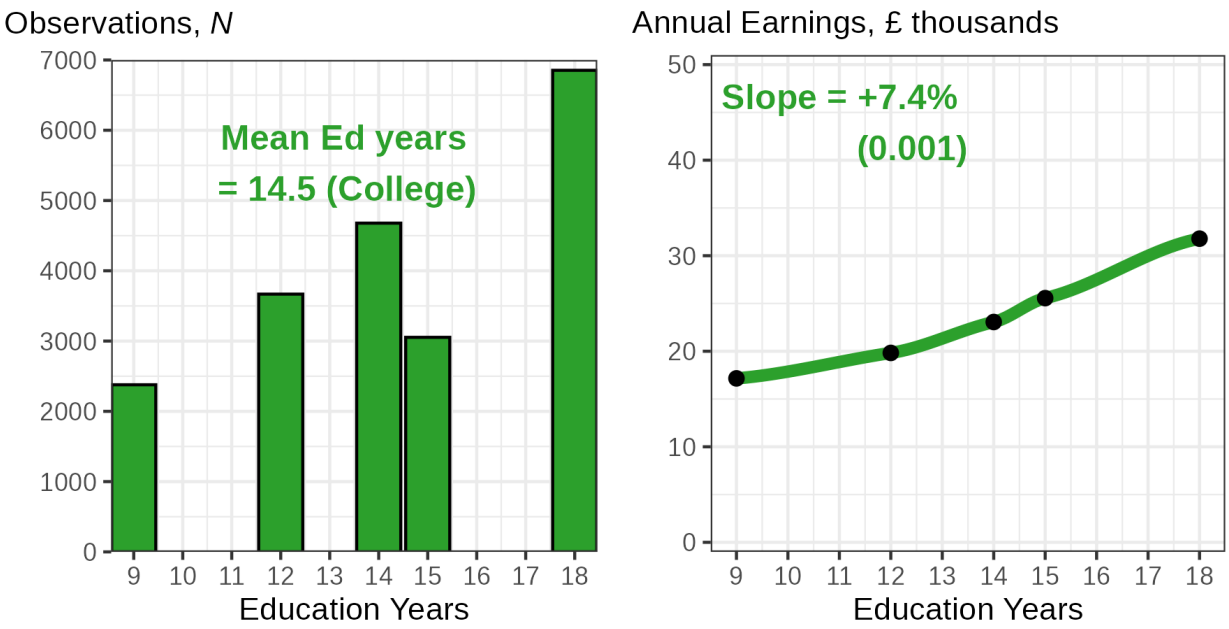
	Analysis sample		Entire sample	
	Mean	SD	Mean	SD
Demographics:				
Male	0.45	0.50	0.46	0.50
Age	53.56	6.48	56.53	8.09
Ethnicity = European	1.00	0.00	0.82	0.39
Any siblings?	1.00	0.00	0.08	0.28
Count of siblings	1.09	0.33	0.09	0.31
Education:				
Ed PGI	0.02	1.00	-0.00	1.00
Ed PGI, parental mean	-0.03	0.75		
Education years	14.54	2.96	14.30	3.22
Qualification, University degree	0.33	0.47	0.33	0.47
Qualification, A-Levels	0.31	0.46	0.27	0.45
Qualification, GCSEs	0.62	0.49	0.52	0.50
Qualification, Professional degree	0.31	0.46	0.29	0.45
Qualification, Vocational degree	0.25	0.43	0.19	0.39
Qualification, No official qualifications	0.12	0.33	0.17	0.38
Labour Market Outcomes:				
Occupation hourly wage, £	19.99	9.02	20.88	9.50
Occupation annual income, thousands £	25.07	16.49	26.56	17.42
Average hours worked, per week	34.73	12.58	35.31	12.70
Household income, $< £18k$	0.10	0.29	0.20	0.40
Household income, $£18 - 31k$	0.22	0.41	0.22	0.41
Household income, $£31 - 52k$	0.30	0.46	0.22	0.42
Household income, $£52 - 100k$	0.24	0.43	0.17	0.38
Household income, $£100k <$	0.06	0.23	0.05	0.21
Observations	20,630		502,129	

Note: This table the Mean and Standard Deviation (SD) of UKB variables, comparing the analysis subsample and the entire UKB sample. The analysis sample refers to individuals with at least one sibling in the UKB data, of European ancestory. All genetic variables are only non-missing among those who report European ancestory.

For the purposes of this study, the primary analysis sample is restricted to those with at least one family member, sibling or parent, also enrolled in the UKB, and non-missing data in occupation code plus education qualifications. The UKB does not personally identify participants, nor their family members, so I follow the genetic literature by identifying

Figure 1: Distribution of UKB Education Measures.

(a) Education Years. **(b)** Education Years + Income.



Note: This figure shows the marginal distribution of education years in the UKB analysis sample, and the correlation between years of education and occupation coded income.

family clusters by measures of genetic similarity.[3] This family-linked subsample enables the examination of within-family variation, allowing for more robust identification strategies that can account for unobserved family-level characteristics that may confound the relationships of interest; see **??**. The presence of multiple family members within the dataset provides a unique opportunity to control for shared genetic and environmental factors that might otherwise bias estimates in standard cross-sectional analyses.

The analysis sample is also restricted to individuals who report to have primarily European ancestry. The genetic measures used in this paper have generally been constructing from primarily white, European descended populations, and have no predictive power out of sample, so are only relevant in this subsample.[4] The only notable differences between the analysis

[3]The design of the UKB allows for a larger number of family clusters than would be expected from a random sample of the British population. This is because study enrolment letters were sent to households, and the household head (or to whomever the letter was addressed) could optionally enrol their family members in addition to themselves. Appendix A.2 describes the family cluster identification procedure in detail, which has been used in previous social science genetic studies (see e.g., Muslimova et al. 2025).

[4]It is an unfortunate fact that systematic genetic measures, such as those considered in this paper, have only been constructed from Biobank samples where the vast majority are white and/or European descended; see Martin et al. (2017) for a further discussion. Note that this implies this paper has nothing to say about race; see Subsection 2.4.

subsample and the entire UKB participant sample is that the analysis sample has more siblings, and is entirely European descended; no other variables are significantly different for the analysis subsample. Table 1 summarises the UKB data, for the analysis sample and among the entire UKB sample. Figure 1 summarises the correlation between education years and occupation coded income.

2.2 Education PolyGenic Index (Ed PGI)

UKB participants also provided saliva samples that were subsequently genotyped using high-density arrays containing over 800,000 genetic markers, producing standardised genetic data files for statistical analysis of the relationship between genetics and social outcomes.

While genetic data provide rich information about individual variation, the relationship between specific genes and social outcomes like education is extraordinarily complex. The biological pathways from DNA to observable traits remain largely opaque — the exact relationship between a single gene, its resulting protein (or its moderating role), and the effects of those proteins are only known for a small number of genes in the human genome. For example, a single gene (and its recessive variant) is known to lead to a protein which does not break down the by-products of alcohol digestion; individuals with this gene generally do not drink alcohol at all because of the simple limit on digestion (Millwood et al. 2023). Complex social behaviours, on the other hand, have no single genes with a known biological pathway, so cannot be summarised in a single gene, nor single biological pathway. Instead complex social behaviours are a result on thousands of genes, and their interactions with the environment.

Rather than attempting to trace all of these intricate biological mechanisms, researchers have developed PolyGenic Indices (PGIs) as dimensional reduction tools that summarise the statistical relationship between thousands of genetic variants and outcomes of interest.[5] This aggregation approach is particularly valuable because individual genetic variants typically have tiny effects on complex traits like educational attainment — often explaining less than 0.01% of the variation. However, by combining information across hundreds of thousands of variants, each contributing a small effect, PGIs can capture meaningful portions of the genetic contribution to these outcomes. These indices are fundamentally correlational measures: they capture statistical associations between genetic variants and outcomes observed in large samples, not well understood biological pathways, and without a clear case for causal

[5] See Uffelmann et al. (2021) for a full explanation of how a GWAS works, and the approach to calculating PGIs.

relationships. The weights used to construct PGIs reflect population-level correlations that may arise through various channels, including biological effects, gene-environment correlations, and population stratification.

The Education PGI (Ed PGI) is a summary measure that aggregates genetic information across thousands of genetic variants into a single index predictive of educational attainment. The construction of the Ed PGI relies on weights derived from a Genome-Wide Association Study (GWAS), which analyse the relationship between genetic variants and education years. In a GWAS, researchers test the association between each Single Nucleotide Polymorphism (SNP) — a one-letter change in the DNA sequence at a specific location — and the outcome of interest, in this case education years. I use the weights calculated in the GWAS conducted by Okbay et al. (2022), which used a sample of over 3 million individuals of European ancestry, primarily from 23&Me data.

The Ed PGI is defined as the weighted sum of whether individual i has the relevant SNP at each DNA location j,

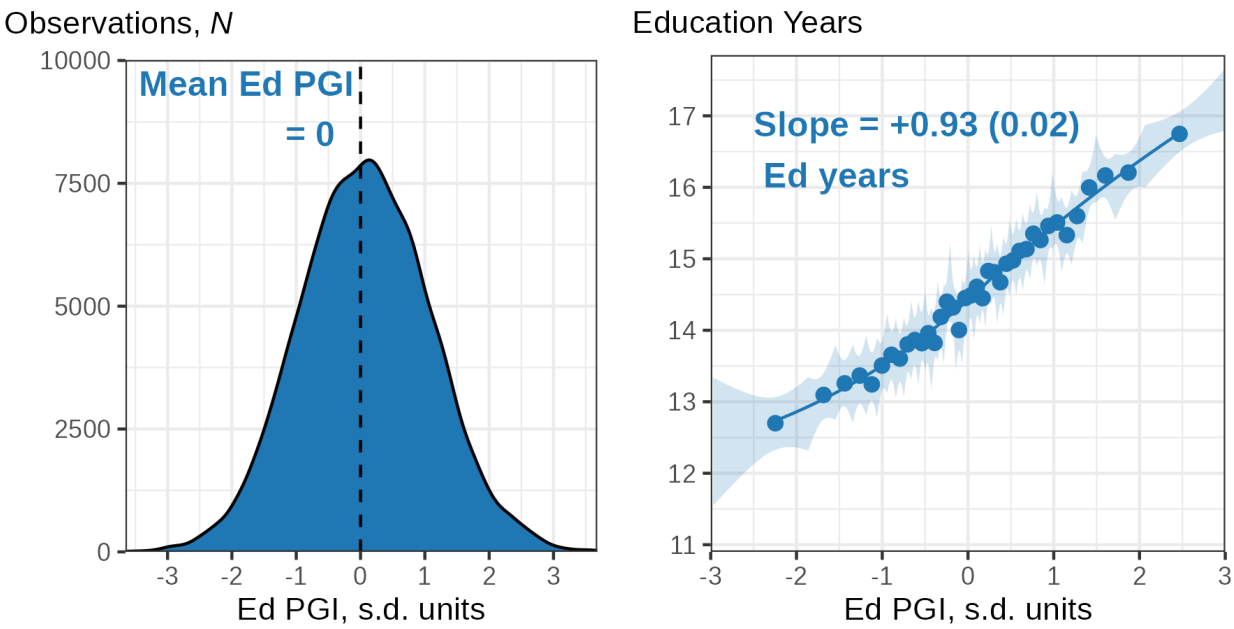
$$Z_i = \sum_{j=1}^{J} w_j \mathbb{1} \left\{ \text{one-letter change at location } j \right\}.$$

w_j are the weights, provided in the supplementary data for Okbay et al. (2022), and I measure the presence of the relevant SNP at each location j in UKB genetic data files. The Okbay et al. (2022) GWAS found that over 8,000 individual SNPs were significant in predicting education years, in a roughly additive manner, after passing a battery of biostatistical tests and finding no evidence for significant interactions. The resulting Ed PGI is standardised to have mean zero and standard deviation one within the entire sample, making it interpretable as the number of standard deviations from the population mean. The Ed PGI captures approximately 12-15% of the variation in educational attainment, making it one of the most predictive PGIs for any complex behavioural trait to date.

The Ed PGI is highly correlated with education years, as it is constructed to be so. A one standard deviation increase in the Ed PGI is associated with 0.93 extra education years (standard error 0.02), with an R^2 value of 0.10 for the raw regression of education years on Ed PGI. It is roughly normally distributed, with standard deviation normalised to 1. It is not surprising that the Ed PGI is normally distributed; it is the weighted sum of thousands of binary variables, among a few thousand individuals, so we should expect the central limit theorem to apply (in the limit) for the Ed PGI. Additionally, it is not surprising that the Ed PGI is correlated with education years; it was literally constructed to be so. The main

Figure 2: Distribution of Education PolyGenic Index (Ed PGI).

(a) Ed PGI. **(b)** Ed PGI + Education Years, Binscatter.



Note: This figure shows the marginal distribution of Ed PGI and education years in the UKB analysis sample. Ed PGI is normalised to have mean 0 and standard deviation 1 among the entire UKB sample; it is roughly normally distributed regardless of this normalisation.

take-away is that the correlation is strong, and has meaningful predictive power for education years. If education had no relationship with genetic variants, then the correlation would be indistinguishable from zero.

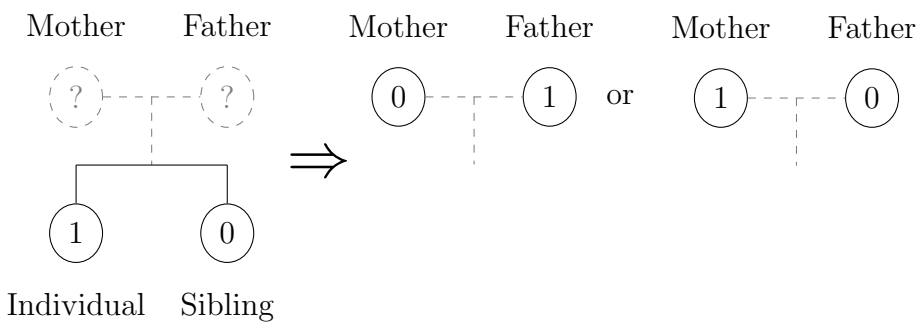
2.3 Imputing Parental PGIs from Siblings

It would be ideal if the UKB contained genetic information for the parents of every UKB participant. However, there are only ≈ 500 participants with a parent also included in the UKB, and only 72 with both parents. To overcome this missing data problem, I use siblings' genetic data to impute the mean parental PGIs from the sample of 20,630 participants with at least one sibling also in the UKB — the sibling analysis sample described above.

I use the Young et al. (2022) procedure, to impute mean parental PGI values using genetic data on participants' siblings. To briefly summarise how this works, consider that each person inherits exactly half their DNA from their mother and half from their father. When we observe two siblings, we can partially reconstruct their parents' genetic information because siblings

both inherit their DNA from parents. Think of it like observing two different samples from the same underlying distribution — the parents' genes. At some genetic locations, siblings inherited the same variant from a parent (like both getting mum's brown-eye gene), whilst at others they inherited different variants (one got mum's brown-eye gene, the other her blue-eye gene). By comparing siblings' genetics across thousands of genetic variants, we can recover about 75% of their parents' combined genetic information — enough to construct a reasonably accurate parental PGI.

Figure 3: Imputing One SNP for Parents from Sibling Data.



Note: This figure illustrates the logic of imputing parental genotypes from sibling data at a single SNP. At each SNP, individuals have a genotype of 0, 1, or 2, representing the number of copies of a particular allele inherited from their parents.

Figure 3 visualises the logic behind imputing parental SNP values based on an individual's, and their sibling(s)'s, genetic data. In the middle panel, the individual has genotype 1 and their sibling has genotype 0. Since the individual must have inherited their 1 allele from one parent and their "0" allele from the other, while the sibling inherited two 0 alleles, we can infer that one parent has genotype 0 (contributing 0 to both children) and the other has genotype 1 (contributing 1 to the individual and 0 to the sibling). The right panel shows an alternative configuration yielding the same sibling genotypes. When parental genotypes cannot be uniquely determined, the imputation uses the expected values based on observed alleles and population frequencies. By applying this logic across hundreds of thousands of SNPs and accounting for segments of DNA shared identically by descent between siblings, we can accurately reconstruct parental PGIs despite never observing the parents directly.

The imputation procedure performs well in the UKB analysis sample. Following the Young et al. (2022) validation approach, I verify the accuracy by examining the small subset of families where parental genotypes are actually observed. The imputed parental PGIs show a correlation of over [X] with true parental PGIs, and regression coefficients are statistically indistinguishable from 1, confirming the imputation is unbiased. Appendix A.3 describes

the full technical detail behind the imputation procedure, and this validation exercise. This high accuracy means I can confidently use the imputed parental PGIs in place of directly observed parental PGI values.

2.4 Ethical Considerations

It is crucial to emphasise that the genetic measures in this study should not be misinterpreted as evidence of genetic determinism. Having a higher Ed PGI does not destine someone to higher education or income; it merely represents only a small statistical tendency, not fate. These tendencies are fundamentally shaped by social policies, institutional structures, and cultural contexts that vary across time and place. Just as eye-sight correcting glasses can eliminate the practical consequences of genetic variation in eyesight, education and economic policies can modify or eliminate the consequences of genetic variation in traits relevant to socioeconomic outcomes.

Furthermore, this research provides no support whatsoever for ideas about "natural" social hierarchies or genetic justifications for inequality. In addition, genetic associations identified within one ancestry group cannot be meaningfully compared across groups due to different genetic architectures, environmental contexts, and historical experiences. Any claim using the measures consider in this paper for inference across racial groups, or different racial groups would be both scientifically invalid, and ethically deficient. The appropriate interpretation of these results is that they highlight how both social and genetic factors influence socioeconomic outcomes, which should inspire greater solidarity and support for policies that promote equality of opportunity rather than acceptance of inequality as natural or inevitable.

3 Random Variation in Genetic Inheritance

Genes are not randomly assigned to children. Genes are inherited from parents, where children receive 13 chromosomes from each parent at conception — half the genes from their mother, half from father.

Parents' traits, such their socioeconomic status, may differ for parents with higher or lower PGI values, confounding association-based estimates of PGI genetic effects among their children.

Papageorge & Thom (2020) document how the Ed PGI predicts earnings, even when controlling for education, and posit that the index contains genes that affect skills related to financial literacy.

The Education PGI is not the only PGI that can be constructed. Following the same methodology, I construct additional PGIs for six health-related traits that may confound the relationship between genetics and socioeconomic outcomes: asthma, bipolar disorder, Body Mass Index (BMI), height, schizophrenia, and type 2 diabetes. Each PGI is calculated using the same approach as the Ed PGI — taking weights from published GWAS conducted on independent populations and computing weighted sums of the relevant genetic variants in the UKB data. These other PGIs serve as important control variables in my analysis for two reasons. First, genetic variants often exhibit pleiotropy, meaning a single variant can influence multiple traits simultaneously. For instance, genetic variants associated with education might also affect mental health conditions, which could independently influence labour market outcomes. Second, population stratification — systematic differences in gene frequencies across ancestry groups — can create spurious correlations between any PGI and outcomes if health conditions vary across these groups.

4 Mendelian Independent Inheritance

Cannot choose who your parents are, and you also cannot choose how much you inherit from them.

Needs at least one cool TikZ figure, showing the design for one genetic variant. This should be a figure showing one copy on each parent's genome leads to a random distribution of children's genetic variant (2, 1, or 0 copies).

Eventually get to the following representation, which is a forumla instrument using the random component as an instrument

$$Z_i^{\text{Child}} - \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_i^{\text{Father}}, Z_i^{\text{Mother}}\right] \quad \longrightarrow \quad \text{Independently assigned.}$$

Figure showing the correlation between Ed PGI and demographics, and other PGIs, then the random component in panel B (which no longer is).

4.1 Addressing Measurement Error in the Ed PGI

Mention that PGI weights are calculated with statistical error, so estimates of their effects are attentuated in un-normalised regressions. This can be ameliorated using two PGIs, with (supposedly) uncorrelated measurement errors, to instrument for eachother in the obviously related framework.

An important econometric consideration is that the weights w_j used to construct the Ed PGI are themselves statistical estimates rather than true population parameters. Although Okbay et al. (2022) estimates these weights from an exceptionally large sample of over 3 million individuals, they remain subject to finite sample sampling variation and estimation error. This measurement error in the weights propagates through to the constructed Ed PGI, resulting in classical attenuation bias when examining correlations between the PGI and outcomes of interest. To address this measurement error problem, I follow Muslimova (2024) and construct a second Ed PGI using alternative weights, also provided by Okbay et al. (2022), that were estimated exclusively from the UK Biobank non-sibling subsample. Crucially, because these two sets of weights were estimated from independent samples, their measurement errors are uncorrelated. This allows me to implement an Obviously Related Instrumental Variables (ORIV) approach, where each PGI serves as an instrument for the other, effectively trading off some statistical precision for reduced attenuation bias. The ORIV estimator provides consistent estimates of the relationship between the true (error-free) Ed PGI and outcomes, correcting for the downward bias that would otherwise affect my results. Appendix **??** provides a detailed exposition of this econometric approach and its implementation in the context of PGIs.

5 Genetic Effects

Section to show the results of the Ed PGI → Education, labour earnings. Clearly define the estimands, and then the estimating equations.

5.1 Genetic Results

This will require using the random components as obviously related instruments for the other PGIs, respectively.

These results will likely be larger than in other papers, because of the measurement error

reducing precision in previous work.

Figure 4: TODO: Make this show the binscatter of Ed PGI → Ed years, and income. Regular colour line for the raw correlation, and yellow line for the random component.

(a) Earnings. **(b)** Ed PGI + Earnings.

Note: This figure shows the marginal distribution of Ed PGI and education years among UKB household heads, in the analysis sample defined above. Ed PGI is normalised within-in sample by the UKB to have mean 0, and standard deviation 1; the distribution shape is unaffected by the normalisation — i.e., it is roughly normally distributed regardless. 3.4% of the analysis sample reports education years less than 10 years (not included in panel b).

TODO: a LaTeX table summarising the results, too.

6 Direct and Indirect Effects

This section gives the conceptual framework for direct and indirect effects.

Take the notation from beforehand, and clearly define the mediation estimands, which are average causal response functions in the mediation settings.

Why is this interesting and different from what has preceded?

6.1 Independent Variation in the Education Channel

Reference my other paper, showing that an informal mechanism analysis is appropriate here. Needs independent variation in the education channel to credibly estimate the mechanisms behind the total genetic effects.

Explain this with a simple Roy model.

Note that this now allows for a choice channel, when previously it had not.

7 Empirical Results

8 Summary and Concluding Remarks

Conclusion goes here.

References

Kweon, H., Burik, C. A., Ning, Y., Ahlskog, R., Xia, C., Abner, E., Bao, Y., Bhatta, L., Faquih, T. O., de Feijter, M. et al. (2025), 'Associations between common genetic variants and income provide insights about the socio-economic health gradient', *Nature human behaviour* pp. 1–12. 2

Martin, A. R., Gignoux, C. R., Walters, R. K., Wojcik, G. L., Neale, B. M., Gravel, S., Daly, M. J., Bustamante, C. D. & Kenny, E. E. (2017), 'Human demographic history impacts genetic risk prediction across diverse populations', *The American Journal of Human Genetics* **100**(4), 635–649. 4

Millwood, I. Y., Im, P. K., Bennett, D., Hariri, P., Yang, L., Du, H., Kartsonaki, C., Lin, K., Yu, C., Chen, Y. et al. (2023), 'Alcohol intake and cause-specific mortality: conventional and genetic evidence in a prospective cohort study of 512 000 adults in china', *The Lancet Public Health* **8**(12), e956–e967. 5

Muslimova, D., Van Kippersluis, H., Rietveld, C. A., Von Hinke, S. & Meddens, S. F. W. (2025), 'Gene-environment complementarity in educational attainment', *Journal of Labor Economics (Forthcoming)* . 4

Okbay, A., Wu, Y., Wang, N., Jayashankar, H., Bennett, M., Nehzati, S. M., Sidorenko, J., Kweon, H., Goldman, G., Gjorgjieva, T. et al. (2022), 'Polygenic prediction of educational attainment within and between families from genome-wide association analyses in 3 million individuals', *Nature genetics* **54**(4), 437–449. 2, 6, 11

Papageorge, N. W. & Thom, K. (2020), 'Genes, education, and labor market outcomes: evidence from the health and retirement study', *Journal of the European Economic Association* **18**(3), 1351–1399. 9

R Core Team (2023), *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria.
URL: *https://www.R-project.org/* 15

Smith, A. (1776), *The Wealth of Nations*, Vol. 1. 1

Sudlow, C., Gallacher, J., Allen, N., Beral, V., Burton, P., Danesh, J., Downey, P., Elliott, P., Green, J., Landray, M. et al. (2015), 'Uk biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age', *PLoS medicine* **12**(3), e1001779. https://doi.org/10.1371/journal.pmed.1001779. 2

Uffelmann, E., Huang, Q. Q., Munung, N. S., De Vries, J., Okada, Y., Martin, A. R., Martin, H. C., Lappalainen, T. & Posthuma, D. (2021), 'Genome-wide association studies', *Nature Reviews Methods Primers* **1**(1), 59. 5

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D.,

Wilke, C., Woo, K. & Yutani, H. (2019), 'Welcome to the tidyverse', *Journal of Open Source Software* **4**(43), 1686. https://doi.org/10.21105/joss.01686. 15

Young, A. I., Nehzati, S. M., Benonisdottir, S., Okbay, A., Jayashankar, H., Lee, C., Cesarini, D., Benjamin, D. J., Turley, P. & Kong, A. (2022), 'Mendelian imputation of parental genotypes improves estimates of direct genetic effects', *Nature genetics* **54**(6), 897–905. 7, 8, 15

A Appendix

This project used computational tools which are fully open-source. Any comments or suggestions may be sent to me at `seh325@cornell.edu`, or raised as an issue on the Github project.

A number of statistical packages, for the R language (R Core Team 2023), made the empirical analysis for this paper possible.

- *Tidyverse* (Wickham et al. 2019) collected tools for data analysis in the R language.

A.1 Occupational Imputed Wages and Income

Describe here the Kweon Koellinger imputation procedure. Reference their supplementary data, and thanks for additional guidance on it.

A.2 Identifying Family Clusters by Genetic Similarity

Describe here.

A.3 Validation of the Young et al. (2022) Imputation

A.4 Two PGIs as Obviously Related Instruments